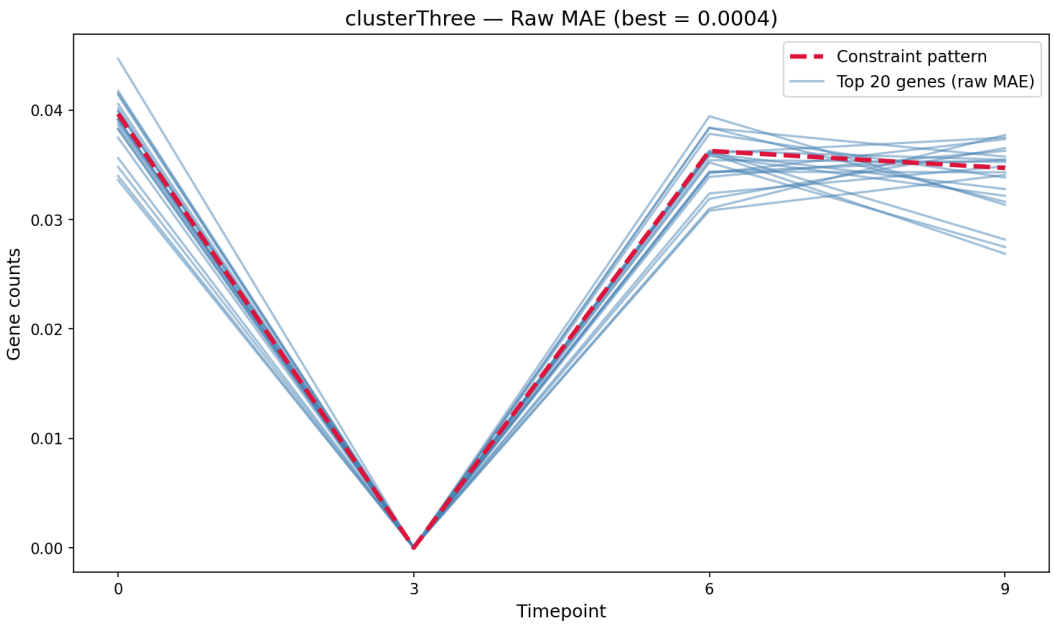
Shape vs. Magnitude

Week of April 23

Binghamton University

April 2026

Where We Left Off (April 16)



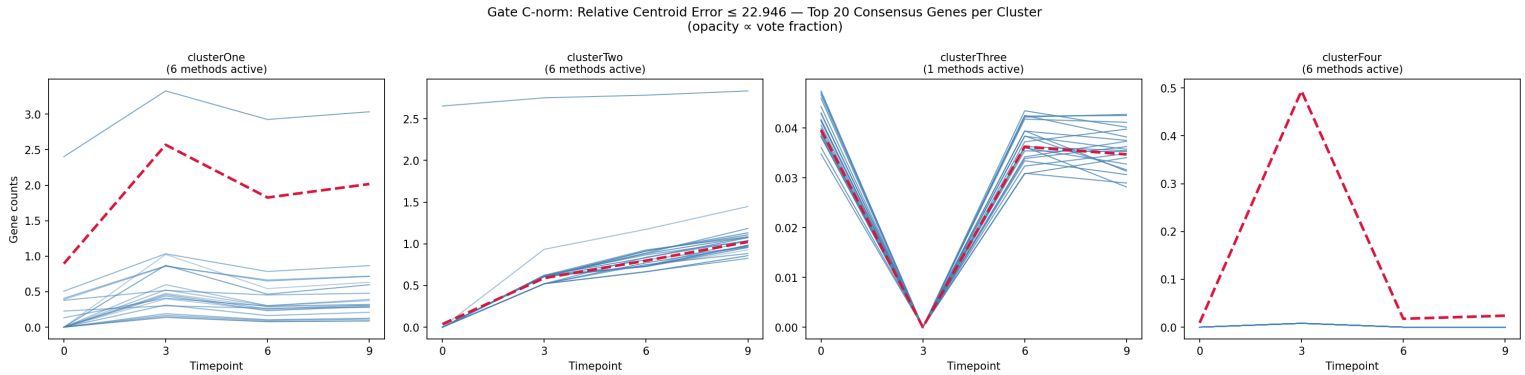
Raw-space MAE gave the best visual fit for clusters Two, Three, and Four.

The six prior methods all operate on *normalized* data — so they find genes with the right *shape* at arbitrary expression levels.

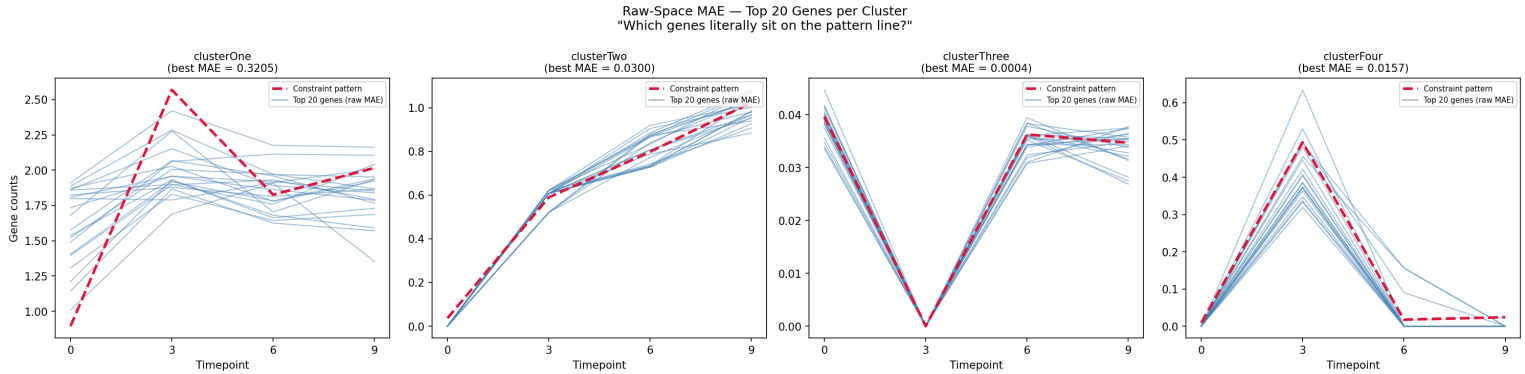
clusterOne was the exception — no genes at the pattern's magnitude also follow its spike shape.

All 4 Clusters — Shape-Based vs. Magnitude-Based

Shape-based gated ensemble



Magnitude-based raw MAE



Question

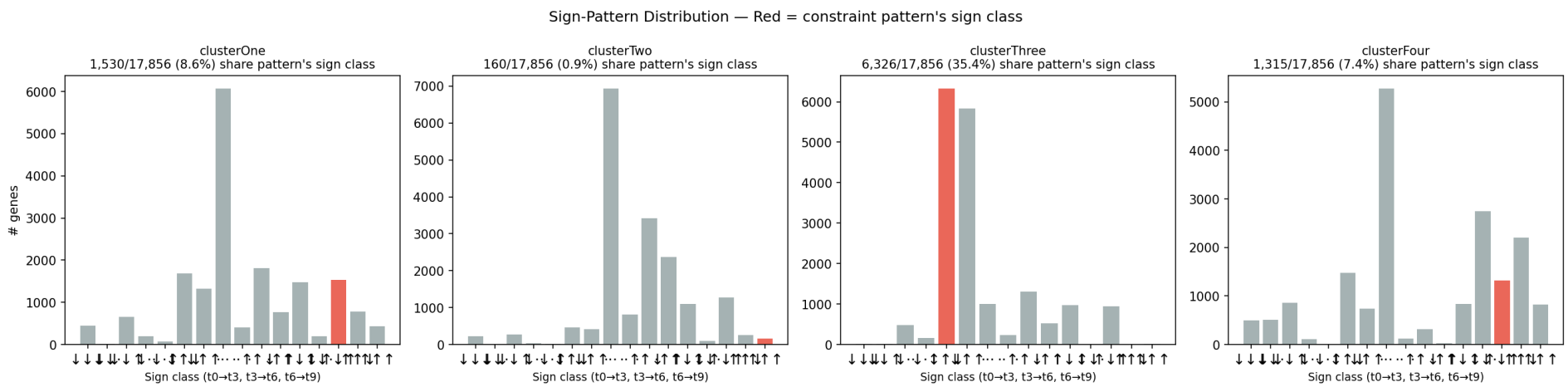
Is biological similarity a matter of

direction of change (shape)

or **absolute expression level** (magnitude)?

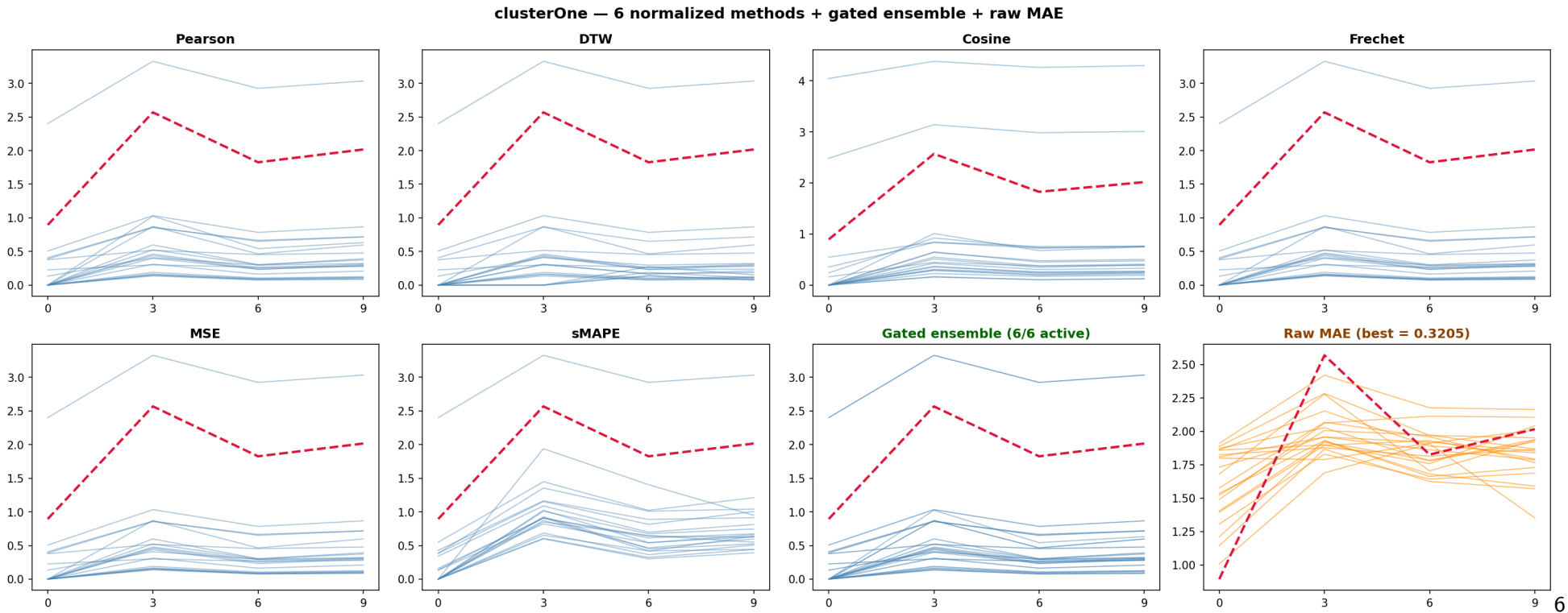
The 4-Timepoint Problem

Each gene's trajectory has 3 transitions ($t_0 \to t_3$, $t_3 \to t_6$, $t_6 \to t_9$). Collapse each to its sign $\Rightarrow$ only $2^3 = 8$ possible "shape classes." All 6 normalized-first methods also erase magnitude, so they're sorting $\sim$17,856 genes into 8 buckets on shape alone. Non-trivial fraction of genes share the constraint's sign class *by accident*.



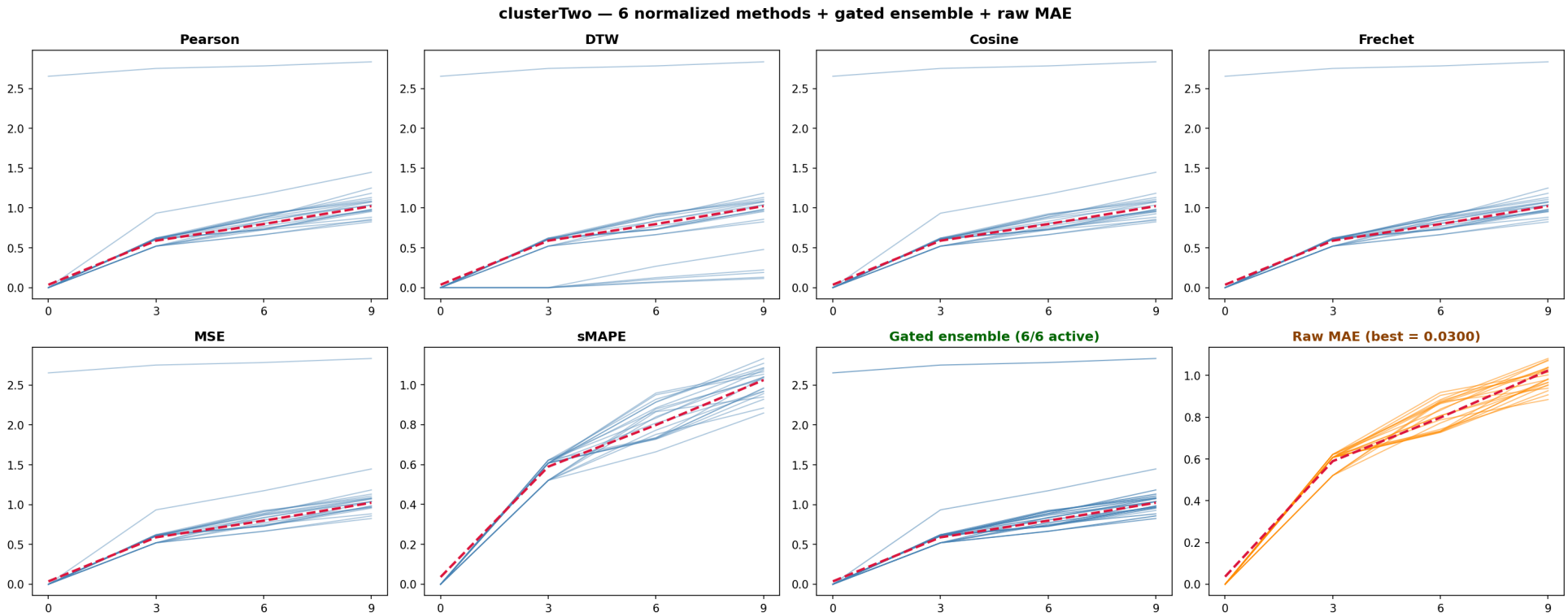
clusterOne — Spike Pattern, Moderate Magnitude

8.6% of genes share the sign class `(+,-,+)`. The 6 normalized methods find spike-like genes mostly *below* the pattern — shape roughly right, magnitude off. Gated ensemble: 6/6 active (no method silenced). Raw MAE: genes sit at the right level but are flat. **Neither path is clean here.**



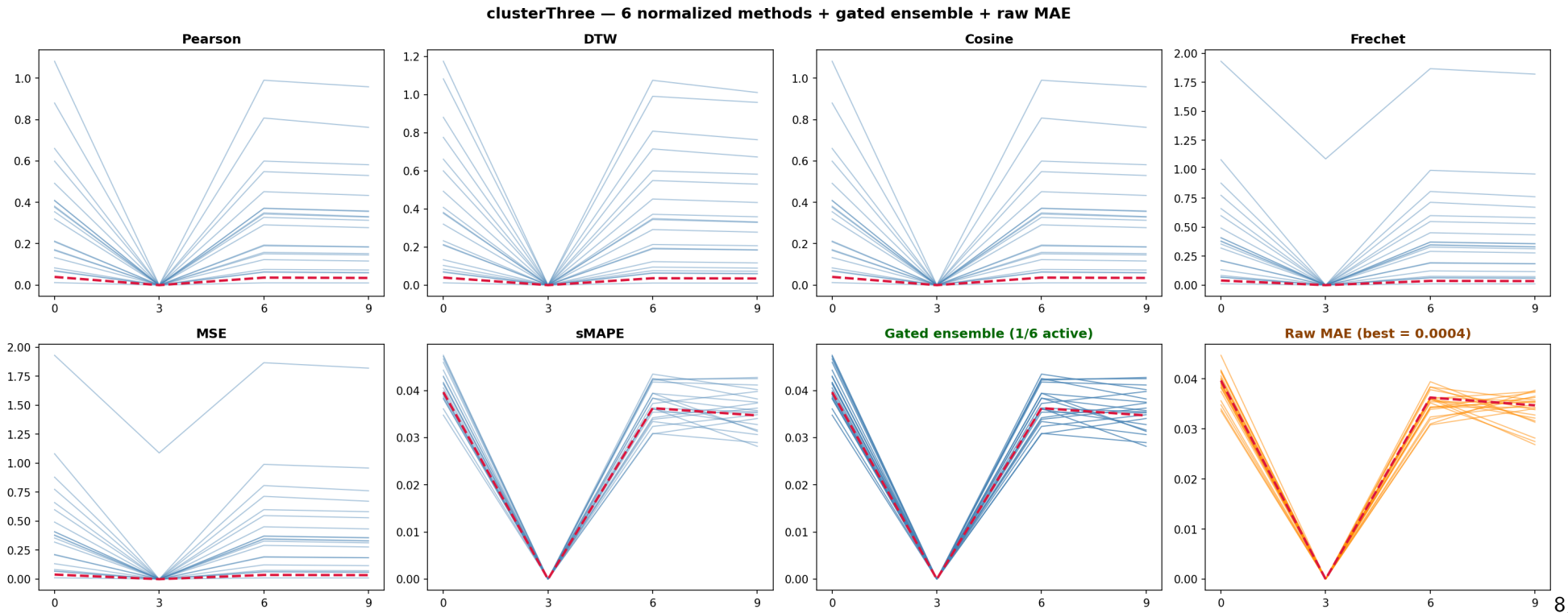
clusterTwo — Monotone Increase

Only 0.9% of genes share (+,+,+) — the rarest sign class, so shape is actually selective here. All 6 methods pick genes that hug both the shape and magnitude. Gated ensemble: 6/6 active, high consensus (RAB14, SYNGR1...). Raw MAE: very tight fit (best ≈ 0.03). **Both paths agree.**



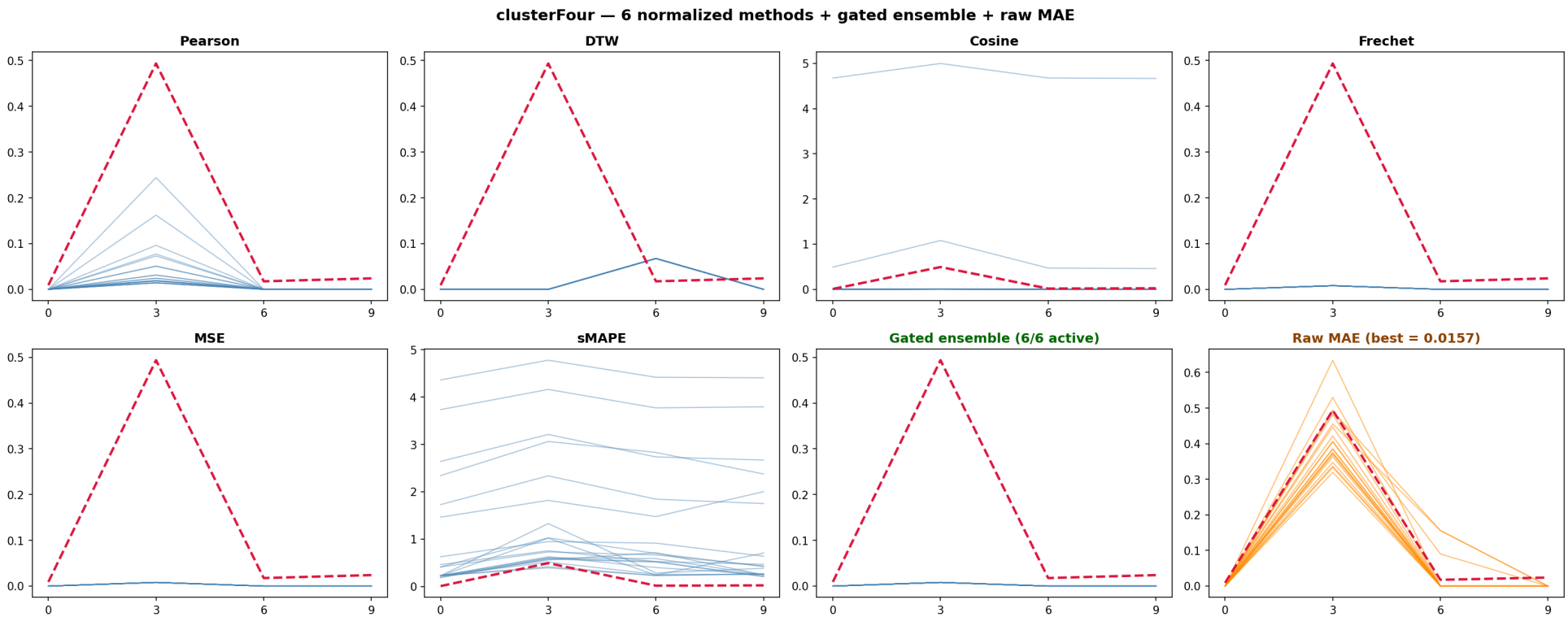
clusterThree — Dip Pattern, Tiny Magnitude

35.4% of genes share `(-,+,-)` — the shape-based filter is weakest here. 5 of 6 normalized methods pick genes 10–40× *above* the pattern (right shape, wrong level). Gated ensemble silences five — only sMAPE survives (1/6 active). Raw MAE: genes sit directly on the pattern. **Magnitude looks closer.**



clusterFour — Sharp Spike, Large Magnitude

7.4% of genes share `(+,-,+)`. 5 of 6 normalized methods pick *flat* genes — normalization erased the spike entirely. Gated ensemble: 6/6 still active — *the gate didn't catch this failure*. Raw MAE captures the spike-shaped genes for the first time. **Magnitude uniquely captures the spike at $t = 3$.**



Question for the Group

What does "similar gene" mean biologically for these TCR constraint patterns?

Direction of change, or absolute expression level?

Appendix

Raw-space methods — no normalization

Appendix — Why Bother?

Only **Pearson** and **Cosine** are scale-invariant by definition. DTW, Fréchet, MSE, and sMAPE are genuine distance metrics — they only *behaved* as shape-based in the April 9 pipeline because every gene was normalized to $(0, 1]$ before they saw it.

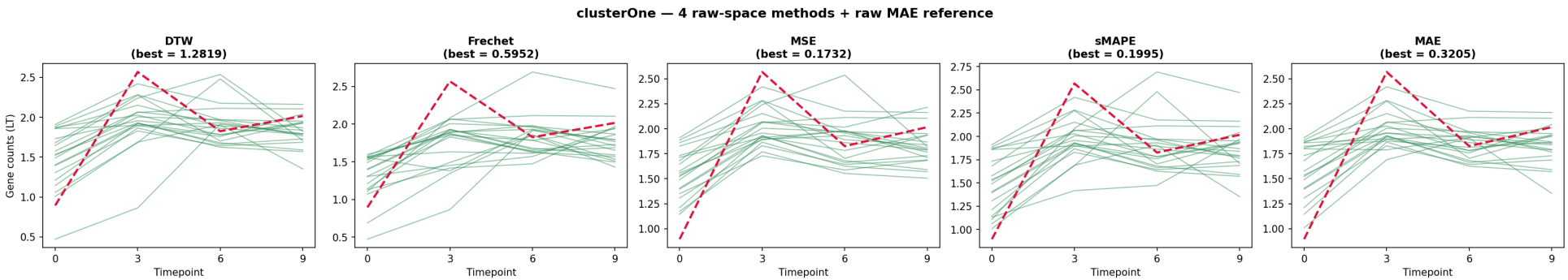
This appendix recomputes those four methods on **raw LT data** (no normalization) to see what their natural picks look like. Raw MAE is included for reference.

Cluster	**DTW**	**Fréchet**	**MSE**	**sMAPE**
clusterOne	17/20	9/20	14/20	17/20
clusterTwo	20/20	17/20	18/20	6/20
clusterThree	20/20	13/20	16/20	17/20
clusterFour	7/20	4/20	17/20	0/20

Top-20 overlap with raw MAE (both computed on unnormalized LT data).

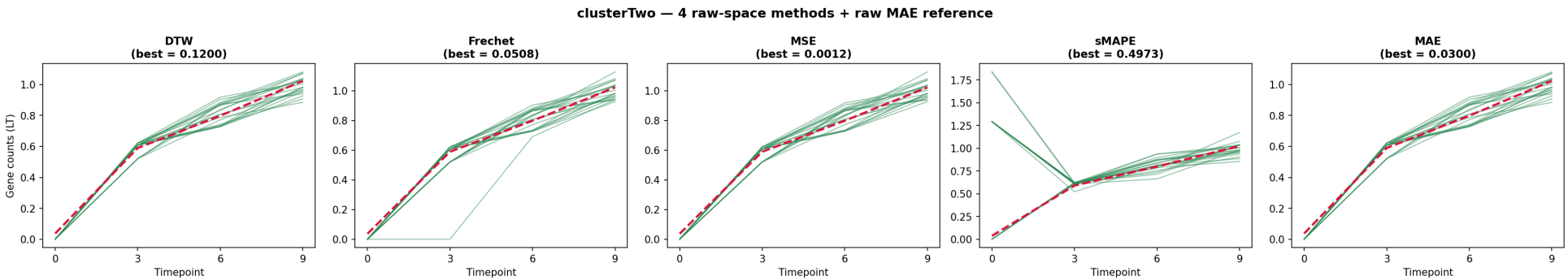
Appendix — clusterOne (raw space)

All four methods find genes close to the pattern. DTW and sMAPE match MAE's picks most tightly (17/20). Fréchet's max-deviation focus picks outliers (9/20).



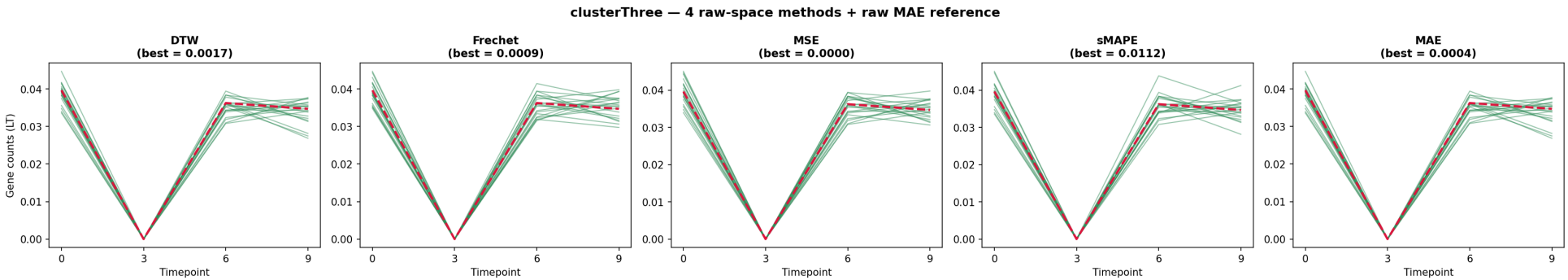
Appendix — clusterTwo (raw space)

DTW is essentially identical to MAE here (20/20). sMAPE is the outlier (6/20) — even on raw data it's structurally relative (divides by $|g| + |p|$), so it ranks genes differently.



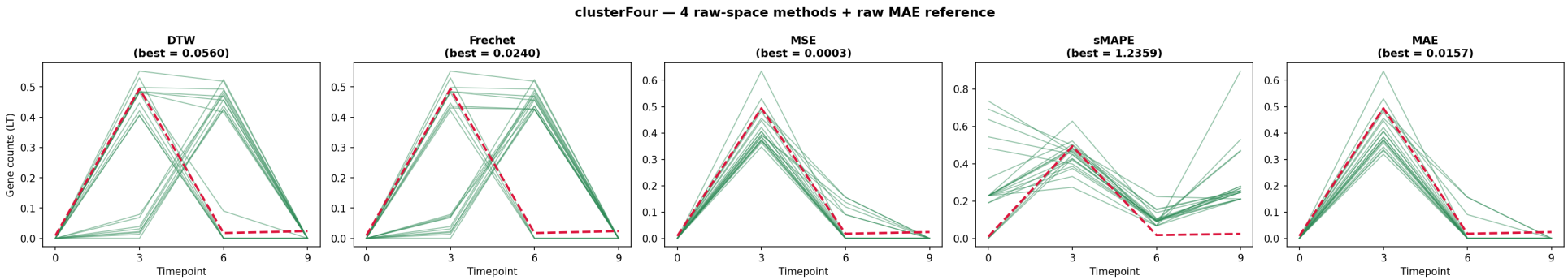
Appendix — clusterThree (raw space)

Without normalization, DTW/Fréchet/MSE/sMAPE all converge on the low-magnitude genes that sit on the pattern. DTW is identical to MAE (20/20). Confirms that magnitude-awareness, not any one metric's machinery, is what drives the correct picks here.



Appendix — clusterFour (raw space)

sMAPE breaks down completely (0/20 overlap with MAE). Pattern values near zero at $t = 6, 9$ make sMAPE's denominator tiny, blowing up its sensitivity. Fréchet (4/20) also struggles on the spike. DTW and MSE behave reasonably.



Appendix — Takeaway

- **DTW and MSE on raw $\approx$ raw MAE.** On 4 timepoints, their extra machinery (temporal warping, squared penalty) doesn't distinguish them meaningfully from pointwise MAE.
- **Fréchet is noisier.** Max-deviation focus punishes one bad timepoint, so it picks different outliers.
- **sMAPE stands apart.** Percentage-style denominator breaks down when pattern values are near zero — erratic across clusters.
- **Practical point:** if magnitude-awareness is what matters, **raw MAE** is the cleanest and most interpretable form of this family.